

AI-Powered Chrome Extension for Real-Time Requirement Analysis

Problem Statement

In software requirement meetings, stakeholders often provide incomplete, vague, or ambiguous information. Analysts usually miss important clarifications during live discussions, leading to poor-quality Functional Requirements (FR) and Non-Functional Requirements (NFR).

Develop an AI-powered Chrome Extension that listens to a live Zoom meeting transcript and generates intelligent clarification questions in real time. Based on the clarification questions and stakeholder responses, the system should generate refined Functional and Non-Functional Requirements.

The system should also evaluate and compare the quality of requirements generated:

- Without clarification support
- With AI-generated clarification support through the Chrome Extension

Objective

The objective of this assignment is to:

- Understand AI-assisted Requirement Engineering in live meetings
- Build a Chrome Extension for real-time transcript analysis
- Detect ambiguous or incomplete stakeholder statements
- Generate intelligent clarification questions automatically
- Refine Functional and Non-Functional Requirements using stakeholder responses
- Evaluate the impact of clarification on requirement quality
- Compare requirement quality before and after clarification

Conversation for Analysis

Hiring Manager:

We need to build an AI-based resume analyzer that can automatically shortlist candidates for our software engineering roles.

ML Engineer:

Okay. How should the system decide which candidates to shortlist?

Hiring Manager:

It should rank them based on relevance to the job description.

ML Engineer:

How are we defining relevance?

Hiring Manager:

Mainly skills and experience. And overall profile strength.

ML Engineer:

What does overall profile strength include?

Hiring Manager:

Things like good companies, solid projects, impactful work.

ML Engineer:

Should we prioritize years of experience?

Hiring Manager:

Yes, but not strictly. Sometimes a strong fresher is better than someone with 5 average years.

ML Engineer:

Do we have historical hiring data to train the system?

Hiring Manager:

We have past resumes and hiring decisions, but they're not very structured.

ML Engineer:

How accurate should the system be?

Hiring Manager:

It should be good enough so that HR trusts it.

ML Engineer:

Do we need explainability? For example, why a candidate was ranked higher?

Hiring Manager:

Yes, that would be useful.

ML Engineer:

Are there any constraints regarding bias or fairness?

Hiring Manager:

Yes, we must avoid bias, especially related to gender or college background.

ML Engineer:

Should the system process resumes in real-time or batch mode?

Hiring Manager:

It shouldn't be slow.

ML Engineer:

What is the expected response time per resume?

Hiring Manager:

Ideally quick.

ML Engineer:

What is the timeline for delivery?

Hiring Manager:

We need an MVP soon.

Features to Implement

- Develop a Chrome Extension for Zoom/meeting transcript monitoring
- Capture or read live meeting transcripts
- Detect vague, incomplete, or ambiguous requirements
- Generate AI-based clarification questions in real time
- Accept stakeholder responses to clarification questions
- Extract Functional Requirements (FR)
- Extract Non-Functional Requirements (NFR)
- Categorize NFRs (Security, Performance, Reliability, Fairness, etc.)
- Generate refined requirements after clarification
- Compare requirements generated:
 - Without clarification
 - With clarification
- Evaluate requirement quality using suitable metrics
- Display generated FRs, NFRs, clarification questions, and evaluation results
- Export generated requirements and evaluation reports as PDF/DOCX/TXT